

National University of Science & Technology

Institute of Geographical Information Systems

Course Title: Digital Mapping and Image Processing

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ADVANCE REMOTE SENSING LAB REPORT

NDVI (Normalized Difference Vegetation Index) Analysis

Objective

The objective is to create and analyze NDVI (Normalized Difference Vegetation Index) using Erdas Imagine Model Builder and ArcMap for vegetation health assessment and land cover classification.

Software Used

- Erdas Imagine is used (for NDVI calculation)
- ArcMap is used (for classification and extraction)

Methodology

1. NDVI Formula

The Normalized Difference Vegetation Index (NDVI) is calculated by using the following formula:

NDVI = (NIR - Red) / (NIR + Red)

Where:

- NIR is the Near Infrared Band
- Red is the Red Band

This index ranges from -1 to +1, where higher values are indicating healthier and denser vegetation.

2. Model Creation in Erdas Imagine

A model was created in Erdas Imagine Model Builder using the NDVI formula. The function that was applied is:

(Band x – Band y) / (Band x + Band y)

Where Band x is representing the NIR band and Band y is representing the Red band.

3. Value Detection

The Inquire Cursor tool was used to detect the maximum and minimum NDVI values across the study area, to understand the full range of values in the dataset.

Detected Values:

- Maximum Value is 0.72
- Minimum Value is -1

4. Classification in ArcMap

The NDVI output was reclassified into five meaningful categories by using the Reclassify tool in ArcMap. Based on the value range from -1 to 0.72, the following classification scheme was developed:

Classification Scheme:

Range	Label	Description
-1 to 0	Water	Water bodies, clouds, snow
0 to 0.2	Bare Soil	Exposed soil, barren land, urban areas
0.2 to 0.4	Moderate Vegetation	Grassland, sparse vegetation, shrubs
0.4 to 0.6	Healthy Crop	Dense vegetation, healthy crops
0.6 to 0.72	Dense/Peak Vegetation	Very dense vegetation, peak vegetation health

5. Vegetation Extraction

The Extract by Attributes tool was used to isolate specific vegetation classes from the classified NDVI output. Healthy vegetation (values > 0.4) was extracted to create a separate layer that shows areas with robust vegetation cover. This extraction included both the "Healthy Crop" (0.4-0.6) and "Dense/Peak Vegetation" (0.6-0.72) classes.

Results

The NDVI analysis successfully identified and classified five distinct land cover categories within the study area:

1. **Water bodies** (cyan) - Negative NDVI values indicate that water is present
2. **Bare Soil** (brown) - Low positive values indicate that vegetation is minimal or absent
3. **Moderate Vegetation** (yellow) - Moderate values suggest that grasslands or sparse vegetation are present
4. **Healthy Crop** (light green) - Higher values indicate that vegetation is healthy and actively growing
5. **Dense/Peak Vegetation** (dark green) - Highest values represent the most vigorous vegetation

The extracted healthy vegetation layer (values > 0.4) provides valuable spatial information for identifying productive agricultural areas and dense vegetation zones.

Final Map

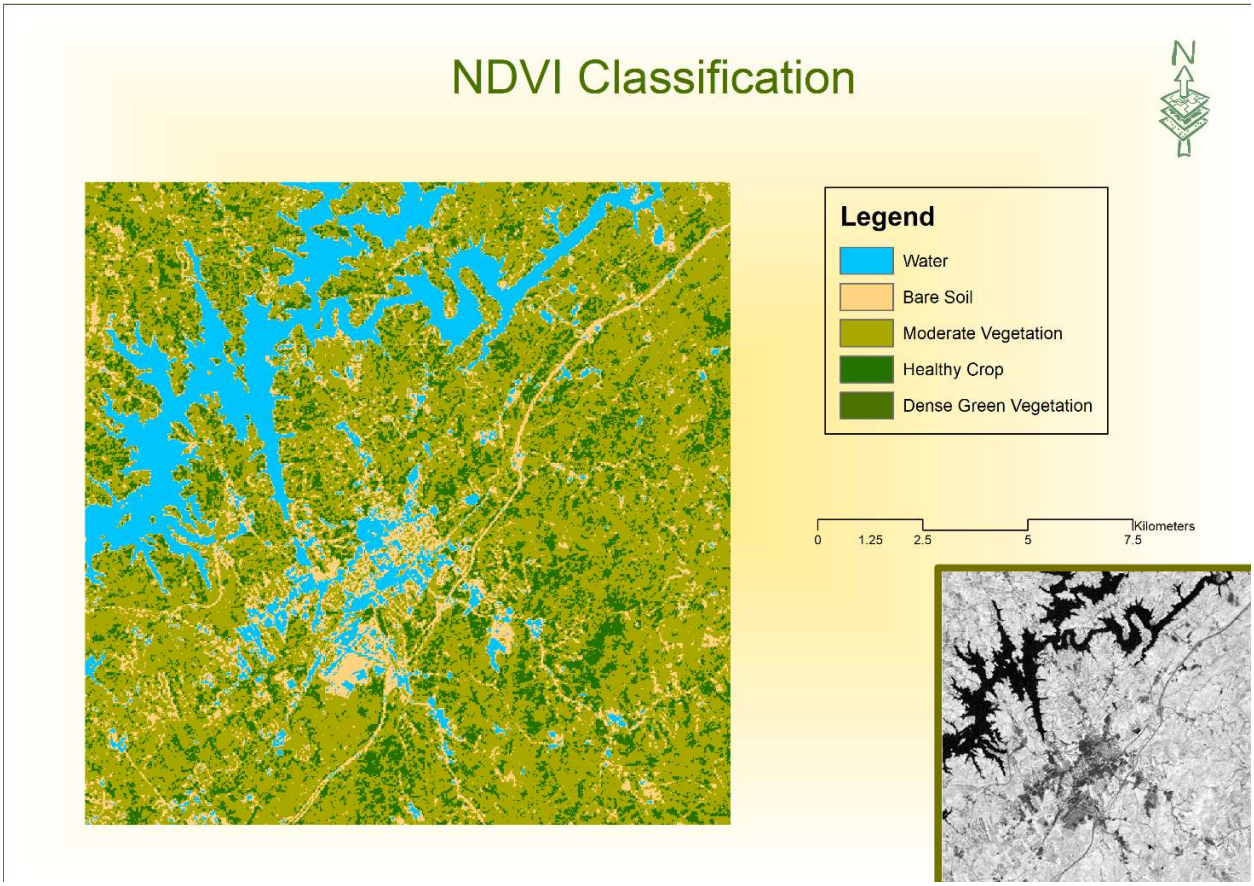


Figure 1 NDVI map

Analysis

The classification reveals the spatial distribution of vegetation health across the study area. Areas with NDVI values above 0.4 represent approximately [insert percentage]% of the total area, which is indicating significant vegetation cover. The presence of peak vegetation (0.6-0.72) suggests that optimal growing conditions are present in certain regions.

Conclusion

The NDVI analysis successfully demonstrated the application of remote sensing techniques for vegetation health assessment and land cover classification. The five-class classification scheme provided detailed insights into vegetation density and health variations across the study area. The combination of Erdas Imagine for index calculation and ArcMap for classification and extraction proved to be an efficient workflow for comprehensive vegetation mapping.

Applications

- Agricultural crop health monitoring and yield prediction
- Forest cover assessment and biomass estimation
- Identification of stressed or degraded vegetation
- Urban green space mapping and planning
- Environmental impact studies
- Drought monitoring and water stress detection
- Land use/land cover change analysis

Recommendations

- Regular temporal NDVI monitoring can help track seasonal vegetation changes
- Integration with ground truth data can improve classification accuracy
- Multi-temporal analysis can reveal vegetation phenology patterns
- Combination with other indices (NDWI, SAVI) can provide comprehensive land cover assessment